FineCL: Exemplar-Free Continual Learning for Fine-Grained Remote Sensing Panoptic Segmentation

Bo Yuan, Danpei Zhao, *Senior Member, IEEE*, Tian Li, Wentao Li, Bowen Yang, and Zhenwei Shi, *Senior Member, IEEE*

***Abstract*—Continual learning (CL) enables the model to continually adapt to new data and tasks. However, the catastrophic forgetting and semantic drift lead to severe model degradation, especially in fine-grained segmentation tasks. Besides, replay-based methods bring extra storage burdens and raise privacy concerns. To address these challenges, we propose FineCL, an exemplar-free continual panoptic segmentation method. Concretely, FineCL proposes an anchoring contrastive distillation (ACD) method to preserve model stability by constraining the semantic consistency. To tackle semantic drift in fine-grained incremental learning (IL), FineCL proposes a dynamic memory bank (DMB) to enhance model plasticity via triplet prototype ranking loss for new knowledge integration. Without storing exemplar data, FineCL sets an uncertainty-guided pseudo-labeling (UGPL) strategy to generate high-confidence pseudo-labels for incremental training. The framework supports end-to-end training with multiple concurrent objectives. Extensive experiments on two fine-grained datasets validate the effectiveness of the proposed model. FineCL outperforms the state-of-the-art method by maximum of 6.6% improvement in long-step continual panoptic segmentation tasks.**

***Index Terms*—Continual learning, Catastrophic forgetting, Knowledge injection, Anchoring contrastive distillation, Panoptic segmentation.**

I. Introduction

CONTINUAL learning (CL), also referred to as incremental learning (IL), breaks off the traditional one-off training manner by enabling models to learn sequentially from nonstationary data streams. Unlike standard supervised learning approaches that operate on static datasets during both training and inference, CL seeks to preserve previously acquired knowledge while incrementally adapting to new information. Accordingly, CL has been widely studied and applied across diverse domains, including remote sensing interpretation [1], [2], autonomous driving [3], [4], medical diagnosis [5], and embodied AI systems [6]. In foundation-model-driven systems, CL is also essential for maintaining long-term adaptability under dynamic tasks [7].

One of the main challenges in CL is *catastrophic forgetting*, where models rapidly lose previously learned knowledge when trained on new tasks or data. This phenomenon occurs when sequential parameter updates overwrite previously learned representations, leading to semantic confusion and performance degradation. While for dense prediction tasks like semantic segmentation and panoptic segmentation, another critical challenge is *semantic drift* [8]. It raises from the evolving composition of the background, which typically contains true background, previously known classes, and emerging unknown classes. Semantic drift manifests as structural knowledge corruption at the pixel level, leading to inconsistent predictions.

According to whether storing old data, current CL methods can be divided into *data-replay* and *data-free* methods [8]. Data-replay methods alleviate forgetting by retaining a subset of past samples. However, they may amplify semantic drift in dense prediction due to biased exemplar selection and limited coverage of intraclass diversity, especially for fine-grained classes and long-tailed instances. Meanwhile, data-free approaches avoid privacy and storage issues, but typically struggle to preserve fine-grained spatial relationships [9]. This limitation is particularly critical for panoptic segmentation, which jointly models *stuff* and *things* and relies heavily on stable feature modeling across incremental steps. Thus, a small amount of semantic drift causes severe performance degradation, especially when fine-grained semantics differ only by subtle appearance variations.

Overall, the obstacles for the continual panoptic segmentation task mainly include: 1) the issue of replay burdens, which fails in data-inaccessible scenarios and 2) the absence of previous knowledge in an exemplar-free manner, leading to severe model degradation in the fine-grained segmentation task.

To address these challenges, we propose FineCL, an exemplar-free continual panoptic segmentation framework. As

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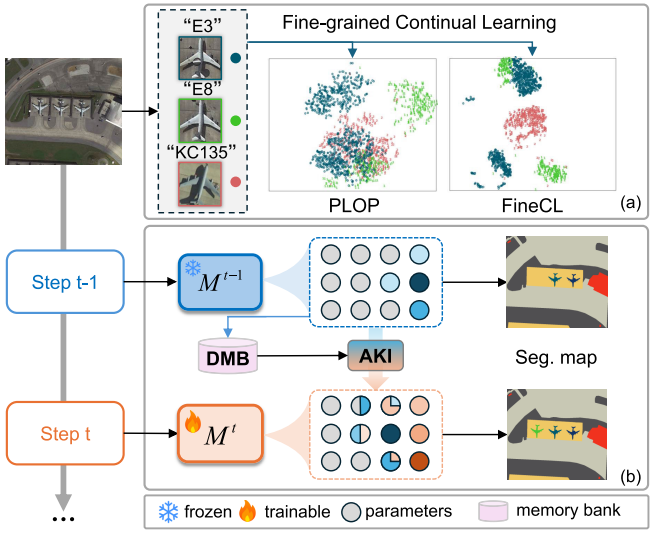


Fig. 1. Overview of FineCL. (a) Illustration of fine-grained semantic drift under CL. (b) Exemplar-free training manner with DMB and AKI to boost the model evolution.

depicted in Fig. 1(a), fine-grained semantics are introduced progressively over incremental steps; the current continual segmentation method often suffers from semantic confusion and cluster collapse, which is aggravated by catastrophic forgetting. In contrast, FineCL better preserves the semantic separation among visually similar classes, maintaining clearer decision boundaries throughout the IL process. As shown in Fig. 1(b), instead of directly storing old data, FineCL maintains a dynamic memory bank (DMB) to preserve learned knowledge as a vector-only multiprototype. At t step, these prototypes serve as compact historical anchors to stabilize the feature space of the current model. Building on DMB, we introduce adaptive knowledge injection (AKI), which retrieves the most relevant prototypes for each location and injects the aggregated memory into dense features via a similarity-conditioned gate. By using Top-K retrieval and a parameter-free gating rule, AKI offers a simple yet effective mechanism to reintroduce historical semantics, which effectively counteracts model degradation in long incremental tasks. Overall, FineCL enables end-to-end training using only the current-step data while effectively retaining fine-grained knowledge over time. Experimental results on two remote sensing benchmarks across eight CL protocols validate the effectiveness of the proposed model.

The main contributions of this article are as follows.

1) We propose an exemplar-free fine-grained continual panoptic segmentation method, addressing both catastrophic forgetting and semantic drift in the remote sensing dense prediction task.
2) We propose a unified CL framework, integrating a lightweight DMB to discard replay burdens and an adaptive knowledge injection to achieve the stability-plasticity balance on fine-grained semantics.
3) Extensive experiments demonstrate that the proposed method achieves state-of-the-art performance on fine-grained continual segmentation tasks, especially in long-horizon CL settings.

II. Related Works

A. Continual Learning

CL has been extensively studied in the context of visual interpretation tasks [10].

Replay-based CL methods mitigate catastrophic forgetting by either storing a subset of past data [11] or generating synthetic samples [12]. Recent work in latent replay [13], [14] reduces storage overhead by caching intermediate features. However, such methods are often constrained by the capacity of generative models and suffer from forgetting phenomena. Another way is to introduce auxiliary data or task [15] to incremental training steps, such as using a web crawler to draw large amounts of data from the Internet [16], [17]. Recent study on generative methods leverages diffusion-based model [12] and VAE-based model [18] to deviate from the distribution of previous tasks.

Replay-free CL methods conduct parameter updates without storing historical data. Techniques including knowledge distillation [19], weight transfer [20], visual prompt tuning [21], and pseudo labeling [22] are employed to tackle the stability and plasticity dilemma. Recent work utilizes LLM to boost the CL using retrieval-augmented generation [23], low-rank adaptation [24], and mixture-of-experts [25].

B. Continual Remote Sensing Image Segmentation

CL in dense prediction can be divided into domain-incremental, class-incremental, and modality-incremental tasks [8].

Domain-incremental methods address the scenario where a model must adapt to new data distributions while maintaining performance on previously encountered domains [26]. Early domain-incremental approaches primarily employed adversarial learning [27]. Recent work has focused on preserving knowledge across multiple domains [26], [28]. Class-incremental methods require models to progressively incorporate new semantic classes while preserving knowledge of previous classes [29]. IDEC [1] proposes a layer-wise knowledge distillation and region-level contrastive learning method to alleviate catastrophic without exemplar memory. LAG [30] proposes an explainable data-limited continual semantic segmentation method. Modality-incremental methods [31], [32] deal with incorporating new data modalities into a model's scope. Recent study on multimodal CL achieves vision–language collaborative incremental updation in a fine-grained remote sensing panoptic perception task [9].

C. Fine-Grained Panoptic Segmentation

Panoptic segmentation aims to unify semantic segmentation of *stuff* regions and instance-level delineation of *things* in a single per-pixel prediction [33]. Fine-grained recognition [34] and panoptic perception [35] investigate the remote sensing interpretation from coarse classes to more detailed subclasses and subtle interclass boundaries.

Existing approaches generally follow two lines. The first line combines a semantic branch for stuff with an instance branch for things, and then applies fusion rules to produce a

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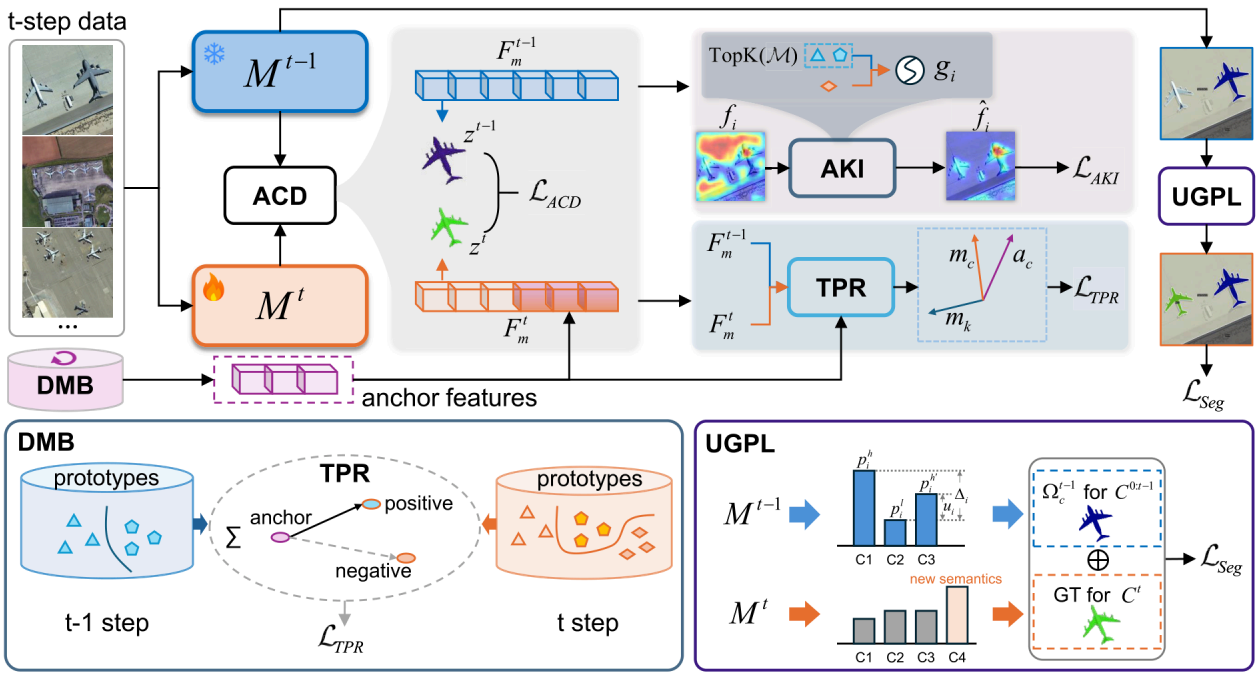


Fig. 2. Pipeline of FineCL. At t step, the proposed model utilizes the frozen M^{t-1} to generate features and pseudo labels to guide M^t training. ACD module constrains the feature consistency on $C^{0:t-1}$ from M^{t-1} and M^t. We maintain a DMB for multiclass prototypes without exemplar replay. AKI injects old knowledge to achieve fine-grained semantics learning. However, UGPL obtains high-quality pseudo labels to boost incremental training.

panoptic map. In remote sensing, these methods incorporate multitask alignment [36], time-sensitive condition [37], and temporal attention [38]. The second line adopts *query-based* frameworks, where a set of learned queries predicts object masks and classes, which naturally aligns with panoptic outputs and can reduce heuristic postprocessing [39].

To address the fine-grained interpretation, recent studies increasingly explore methods including referring segmentation [40]. Some work leverages foundation models [41], [42] to provide better generalization and semantic alignment. Another direction introduces contrastive objectives to compact intraclass embeddings and enlarge interclass margins [43]. Despite these advances, fine-grained panoptic segmentation remains challenged by expensive annotation, domain shifts, and insufficient interpretation. These limitations motivate research on more data-efficient learning, robust cross-domain adaptation, and continual updating mechanisms [9] that can accumulate fine-grained knowledge during incremental steps.

III. Preliminaries

The proposed continual panoptic segmentation method conducts a sequential learning scheme. Let $\mathcal{D} = \{(x_i, y_i)\}$ signify the training dataset, where $x_i \in \mathbb{R}^{C\times H\times W}$ denotes the original image and $y_i \in \mathbb{Z}^{H\times W}$ as the annotation. The CL training procedure consists of $\{Step^0, Step^1, \ldots, Step^T\}$, where T is the numbers of CL steps. Using M^{t-1} and M^t represent the model trained at $Step^{t-1}$ and $Step^t$, respectively.

IV. Method

In this article, we propose FineCL, an exemplar-free continual panoptic segmentation framework. As depicted in Fig. 2, FineCL contains an anchoring contrastive distillation (ACD) embedded in the pixel decoder, a DMB that stores class-wise multiprototypes as compact historical knowledge, and an AKI module that retrieves and injects only the most relevant historical knowledge with similarity-aware gating to avoid interference on irrelevant semantics. In addition, we introduce an uncertainty-guided pseudolabeling method to boost the incremental training.

A. Anchoring Contrastive Distillation

Following [44], the encoder consists of a hierarchical backbone that extracts multistage features and a pixel decoder. Given an input image $I \in \mathbb{R}^{H\times W\times C}$, the encoder produces a set of multiscale feature maps $\{F_m\}$, where $F_m \in \mathbb{R}^{H_m \times W_m \times C_m}$. The pixel decoder then fuses these multistage features using a transformer-style encoder with multiscale deformable attention.

At incremental step t, we train the current model M^t and maintain a frozen M^{t-1} copied from $t-1$ step. To explicitly stabilize the feature space, we propose to use M^{t-1} to generate anchor features.

Instead of enforcing strict pixel-wise feature matching, we enforce instance-level consistency for fine-grained thing classes as an anti-forgetting prior. For fine-grained instance masks, let $\{\mathcal{R}_k\}_{k=1}^{K_I}$ be old-class instance regions on image I, where each $\mathcal{R}_k$ corresponds to one predicted instance mask. For a chosen decoder feature map F_m, we compute instance embeddings by masked average pooling

$$z_k^{t-1} = \Psi\left(\frac{1}{|\mathcal{R}_k|}\sum_{i\in\mathcal{R}_k} W_m F_m^{t-1}(i)\right)$$
$$z_k^{t} = \Psi\left(\frac{1}{|\mathcal{R}_k|}\sum_{i\in\mathcal{R}_k} W_m F_m^{t}(i)\right) \quad (1)$$

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where $\Psi(\cdot)$ denotes ℓ_2 normalization. $W_m \in \mathbb{R}^{C' \times C_m}$ is a 1×1 conv projecting features into a shared embedding space. We then define fine-grained instance consistency via ACD

$$\mathcal{L}_{ACD} = \frac{1}{K_I} \sum_{k=1}^{K_I} \left(1 - cos\left(z_k^{t-1}, z_k^t\right)\right) \tag{2}$$

where $\mathcal{L}_{ACD}$ encourages M^t to preserve fine-grained instance representations on instances belonging to previously learned classes $C^{0:t-1}$, thereby reducing representation drift without exemplar replay.

This exemplar-free anchor guidance acts as a stability prior at the t step; it constrains representation drift on old-class regions while the model learns new semantics.

B. Uncertainty-Guided Pseudo Labeling

Since ground-truth labels for old classes $C^{0:t-1}$ are unavailable at t step, we use M^{t-1} to produce pseudo labels and retain only high-confidence pixels via uncertainty constraint. For dense CL tasks, the pixel-wise prediction needs pixel-wise optimization rather than class-wise threshold since the large intraclass variance. We propose an uncertainty-guided pseudo-labeling (UGPL) method as follows:

$$\Omega_c^{t-1} = \left\{ i | \text{argmax } p_i^{t-1} = c, \max\ p_i^{t-1} \geq \frac{u_i}{\Delta_i} \right\} \tag{3}$$

where Ω_c^{t-1} denotes the pseudo labels for class c generated from M^{t-1}. p^{t-1} denotes class probabilities from M^{t-1} at t step. u_i is the uncertainty and Δ_i is the uncertainty range of pixel at i, which are calculated by

$$u_i = 1 - \left| p_i^h - p_i^{h'} \right| \tag{4}$$

$$\Delta_i = \left| p_i^h - p_i^l \right| \tag{5}$$

where p_i^h, $p_i^{h'}$, and p_i^l denote the highest, second-highest, and lowest prediction scores of pixel i. Intuitively, confident predictions yield small u_i and large Δ_i. Notably, u_i/Δ_i serves as a pixel-adaptive confidence criterion derived from the prediction distribution, rather than a manually designed global threshold. We mainly focus on how such adaptive uncertainty-guided filtering improves the reliability of pseudo labels in dense CL.

To suppress pixel-level noise and obtain stable class-wise representations for memory updates, we compute a region embedding for each class c by masked average pooling

$$r_c = \frac{1}{|\Omega_c|} \sum_{i \in \Omega_c} f_i \tag{6}$$

where f_i denotes the feature from M^t at pixel i. The class embedding r_c provides more compact and robust semantics for prototype update.

C. Dynamic Memory Bank

We maintain a DMB containing class-wise prototypes for previously learned classes to achieve exemplar-free CL. At t step, we store a set of prototypes for each historical class $c \in C^{0:t-1}$

$$\mathcal{M}_c = \left\{ m_{c,k} \in \mathbb{R}^{C_p} \mid k = 1, \ldots, K \right\} \tag{7}$$

where K is the number of prototypes per class and C_p is the prototype embedding dimension. The full memory bank is $\mathcal{M} = \cup_c \mathcal{M}_c$.

For each region embedding $m_{c,k}$, its assigned prototype is treated as positive, and the nearest prototypes of other classes are treated as negatives. We extract anchor embedding a_c for class c by masked pooling feature F_m^{t-1}

$$a_c = \Psi\left(\frac{1}{\left|\Omega_c^{t-1}\right|} \sum_{i \in \Omega_c^{t-1}} W F_m^{t-1}(i) \right) \tag{8}$$

where W is a 1×1 projection into the prototype space and $\Psi(\cdot)$ denotes ℓ_2 normalization.

When learning in incremental steps, the prototypes are updated by accommodating new semantics while maintaining the old semantics. To improve anti-forgetting ability, we propose a triplet prototype ranking (TPR) objective. For each old class $c \in C^{0:t-1}$, the derived anchor should score all prototypes of the same class higher than prototypes from other classes. Given an anchor a_c, we encourage it to assign high similarity to prototypes of its own class and low similarity to all other prototypes. We define a temperature-scaled similarity score using cosine similarity and optimize the following ranking loss:

$$\mathcal{L}_{TPR} = -\sum_{c \in C^{0:t-1}} log \frac{\sum_{m \in \mathcal{M}_c} \exp\left(cos\left(a_c, m\right)/\tau_r\right)}{\sum_{m \in \mathcal{M}} \exp\left(cos\left(a_c, m\right)/\tau_r\right)} \tag{9}$$

where τ_r is a temperature controlling the sharpness of the ranking distribution. The numerator aggregates embeddings from all same-class prototypes, while the denominator normalizes over the entire memory bank. Notably, the TPR loss is applied to a DMB inherited from previous incremental steps.

At each step, we update prototypes using the class embedding r_c computed from current data. For each class c, we assign r_c to the closest prototype in $\mathcal{M}_c$

$$k* = \underset{k}{\text{argmax}} \cos\left(r_c, m_{c,k}\right) \tag{10}$$

$$m_{c,k*} \leftarrow (1 - \eta)\, m_{c,k*} + \eta \Psi\left(r_c\right) \tag{11}$$

where η is the update rate. Notably, the DMB stores only vectors and is updated without storing exemplars. For a newly introduced class that is not stored in the DMB, we initialize its prototype memory using the current-step class embedding, then apply the prototype assignment and update rule.

D. Adaptive Knowledge Injection

For dense panoptic segmentation, directly constraining the model via global distillation is still insufficient for fine-grained semantics, which may introduce feature shifts. We introduce AKI, which retrieves the most relevant historical prototypes from DMB via class-wise multiprototypes and injects them into the current step features through a gating mechanism.

1) Prototype Retrieval: At t step, given a feature map F_m^t, we flatten spatial locations and denote the feature at location i as $f_i \in \mathbb{R}^{C_m}, i \in \{1, \ldots, H_m W_m\}$. We use two lightweight projections, including $W_q \in \mathbb{R}^{C_p \times C_m}$ to map features to the prototype space and $W_v \in \mathbb{R}^{C_m \times C_q}$ to map retrieved prototypes back to feature space.

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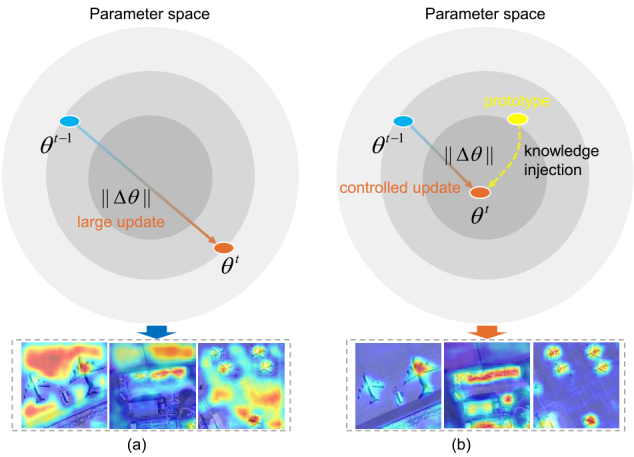


Fig. 3. Effect of AKI on parameter updates. By injecting retrieved prior knowledge, AKI selectively strengthens fine-grained semantic cues while stabilizing previously learned representations. (a) FT. (b) FineCL.

We first project and normalize each feature into the prototype space

$$q_i = \Psi\left(W_q f_i\right) \in \mathbb{R}^{C_p} \tag{12}$$

where $\Psi(\cdot)$ denotes ℓ_2 norm. For each prototype $m_j \in \mathcal{M}$, we compute cosine similarity $s_{i,j} = \cos(q_i, \Psi(m_j))$. To avoid injecting irrelevant knowledge, we select the Top-K most similar prototypes $N_i = \text{TopK}(\{s_{i,j}\}_{j=1}^{|\mathcal{M}|})$ and then aggregate these prototypes using a temperature-controlled softmax

$$\alpha_{i,j} = \frac{\exp\left(s_{i,j}/\tau\right)}{\sum_{k\in N_i}\exp\left(s_{i,k}/\tau\right)}, \quad j \in N_i \tag{13}$$

$$\widetilde{m}_i = \sum_{j\in N_i} \alpha_{i,j}\Psi\left(m_j\right) \in \mathbb{R}^{C_p} \tag{14}$$

where τ controls the sharpness of the retrieval distribution.

2) Residual Gated Injection: We design a lightweight gate based on the residual format, ensuring injection is strong only when retrieved prototypes are truly relevant while maintaining the semantic integrity. Concretely, we define the gate as

$$g_i = \sigma\left(\mathcal{K}\cdot\max_{j\in N_i} s_{i,j}\right) \in (0,1) \tag{15}$$

where $\sigma(\cdot)$ is the Sigmoid function and $\mathcal{K}$ controls the gate sensitivity. Then, we update the feature by a residual injection

$$\hat{f}_i = f_i + g_i \cdot W_v\widetilde{m}_i \tag{16}$$

where W_v maps memory vectors back to the feature dimension. $\hat{f}_i$ denotes the feature after injection.

We apply AKI consistency on pseudo-labeled old regions Ω^{t-1} to stabilize features for previously learned semantics. Thus, the optimization objective of AKI is defined as

$$\mathcal{L}_{AKI} = \frac{1}{\left|\Omega^{t-1}\right|}\sum_{i\in\Omega^{t-1}}\left|\left|\Psi\left(W_c\hat{f}_i^t\right) - \Psi\left(W_c f_i^{t-1}\right)\right|\right|_2^2 \tag{17}$$

where $W_c \in \mathbb{R}^{C_c\times C_m}$ is a shared projection for dimension alignment and $\hat{f}_i^t$ is the injected feature.

As illustrated in Fig. 3, AKI reuses historical prototypes to stabilize dense representations. By combining Top-K prototype retrieval with gated residual injection, AKI improves retention of old classes via feature-level guidance to induce controlled parameter update.

E. Overall Objective

The proposed model enables end-to-end training via multiple objectives. It consists of the segmentation loss $\mathcal{L}_{Seg}$ via cross-entropy-based supervision, the ACD loss $\mathcal{L}_{ACD}$ to stabilize fine-grained instance representations across incremental steps, the triplet prototype ranking loss $\mathcal{L}_{TPR}$, and the knowledge injection loss $\mathcal{L}_{AKI}$. The overall objective can be formulated as

$$\mathcal{L} = \mathcal{L}_{Seg} + \mathcal{L}_{ACD} + \mathcal{L}_{TPR} + \lambda\mathcal{L}_{AKI} \tag{18}$$

where λ controls the strength of knowledge injection and balances stability against plasticity.

V. Experiments

A. Datasets and Protocols

1) Datasets: We employ two datasets including FineGrip [35] and DroneDeploy [45] to validate the proposed model. FineGrip is a fine-grained dataset that contains 25 semantic classes. FineGrip consists of 2649 images, of which 901 are for training and 1748 are for validation. Specifically, the dataset contains 12 054 fine instance segmentation masks, while the background class includes 7599 semantic masks. DroneDeploy consists of a large number of aerial images which has a spatial resolution of 0.1 meter. It contains seven semantic categories, including *Building*, *Clutter*, *Vegetation*, *Water*, *Ground*, *Car*, and *Ignore*. The DroneDeploy dataset contains a total of 55 image samples. Due to the large size of the original image samples and the presence of a large amount of background areas, this article crops the images as patches with 512 $\times$ 512 pixels. Eventually, 2382 image samples were obtained, among which 1882 were for the training set and 500 for the validation set.

2) Protocols: Following [8], we conduct experiments on few-step with few classes (FSFC), multistep with few classes (MSFC), and multistep with multiclasses (MSMCs) settings. Particularly, there are four incremental tasks on FineGrip, including 20-5 (two steps), 15-5 (three steps), 15-2 (six steps), and 10-5 (four steps). Specifically, we initialize training with all *stuff* classes to establish stable background semantics. For DroneDeploy, 6-1 (two steps), 4-3 (two steps), 4-1 (four steps), and 2-2-2-1 (four steps) are performed.

3) Metric: For panoptic segmentation, we adopt panoptic quality (PQ) as the primary evaluation metric, which jointly reflects recognition accuracy and segmentation quality. For semantic segmentation, we report mean intersection over union (mIoU) as the standard performance measure. Particularly, we report the performance of the initial classes and the incremental classes, as well as all classes, to evaluate both the stability and plasticity.

B. Implementation Details

We use Mask2Former [44] as a baseline. The training epochs are 50 for all incremental steps. At each incremental

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TABLE I

QUANTITATIVE COMPARISON ON FINEGRIP IN TERMS OF PQ (%). THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	20-5 (2 steps)			15-5 (3 steps)			15-2 (6 steps)			10-5 (4 steps)		
	$C^{1:15\&st}$	$C^{16:20}$	all	$C^{1:10\&st}$	$C^{11:20}$	all	$C^{1:10\&st}$	$C^{11:20}$	all	$C^{1:5\&st}$	$C^{6:20}$	all
fine tuning	1.01	4.22	1.65	2.85	5.11	3.75	1.28	0.77	1.08	0.67	0.49	0.56
MiB [46]	13.75	20.53	15.11	16.38	18.26	17.13	10.49	8.93	9.87	12.05	10.90	11.36
PLOP [47]	24.89	19.09	23.73	24.21	18.33	21.86	11.85	7.21	9.99	12.88	10.47	11.43
RCIL [48]	24.12	23.67	24.03	23.56	22.90	23.30	12.15	7.13	10.14	14.52	11.26	12.56
IDEC [1]	27.85	31.44	28.57	26.80	25.12	26.13	14.25	7.82	11.68	16.23	12.09	13.75
LAG [30]	27.06	32.59	28.16	29.71	25.41	27.99	17.20	10.30	14.44	16.29	11.38	13.34
CPP [9]	27.50	33.11	28.62	30.17	**25.84**	28.44	17.73	10.62	14.89	18.55	**14.79**	**16.29**
FineCL	**28.35**	**34.22**	**29.52**	**31.26**	25.53	**28.97**	**18.29**	**12.25**	**15.87**	**20.48**	13.15	16.08
offline	57.30	53.27	56.49	58.85	52.95	56.49	58.85	52.95	56.49	55.20	57.35	56.49

TABLE II

QUANTITATIVE COMPARISON ON DRONEDEPLOY IN TERMS OF MIOU (%). THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	6-1 (2 steps)			4-3 (2 steps)			4-1 (4 steps)			2-2-2-1 (4 steps)		
	$C^{1:6}$	C^{7}	all	$C^{1:4}$	$C^{5:7}$	all	$C^{1:4}$	$C^{5:7}$	all	$C^{1:2}$	$C^{3:7}$	all
fine tuning	3.32	14.79	5.79	5.33	11.21	7.85	1.01	3.22	1.96	1.35	2.79	2.38
MiB [46]	55.23	28.75	51.45	49.41	34.29	42.93	44.11	24.38	35.65	36.53	18.95	23.97
PLOP [47]	61.89	29.51	57.26	51.22	41.10	46.88	46.52	28.40	38.75	41.64	24.83	29.63
RCIL [48]	61.55	34.19	57.64	52.11	48.45	50.54	45.29	31.52	39.39	41.53	25.90	30.37
IDEC [1]	**65.37**	38.12	**61.47**	**60.82**	54.49	58.11	55.19	36.64	47.24	54.76	31.88	38.42
LAG [30]	64.92	37.35	60.98	60.52	53.27	57.41	56.01	36.50	47.65	55.62	31.78	38.59
CPP [9]	64.28	38.74	60.63	59.55	**56.22**	58.12	56.27	37.01	48.02	**56.25**	32.21	39.08
FineCL	65.15	**39.25**	61.45	60.69	55.21	**58.34**	**56.32**	**37.26**	**48.15**	56.21	**33.52**	**40.00**
offline	69.75	46.29	66.40	67.22	65.08	66.30	67.22	65.08	66.30	69.49	65.56	66.68

step, the learning rate was initialized as 0.0001 and decayed to 0.00001 through multistep decay. For fair comparison, the backbone and pixel decoder are initialized from the previous step and then updated end-to-end together with the segmentation heads. All experiments are conducted on a workstation with two NVIDIA A800 GPUs.

C. Quantitative Analysis

We validate the proposed with up-to-date methods including MiB [46], PLOP [47], RCIL [48], IDEC [1], LAG [30], and CPP [9] on two datasets. Notably, we report the model performance under *fine tuning* and *offline* settings. Specifically, *fine-tuning* denotes sequential training without any anti-forgetting strategy, reflecting the extent of catastrophic forgetting under naive updates, while *offline* refers to jointly training on all classes with access to the full dataset, serving as an approximate upper bound.

1) FineGrip: For the continual panoptic segmentation task, we evaluate the PQ of the initial classes and the incremental classes after the final incremental step. Compared with conventional class-incremental settings, FineGrip is more difficult because many foreground classes have similar appearances and differ only in subtle semantic details, making CL more susceptible to semantic drift and interclass confusion. Thus, we initialize the CL with stuff classes to emphasize the fine-grained CL on foreground semantics. As shown in Table I, the fine-tuning method suffers from severe catastrophic forgetting and semantic chaos, leading to drastic performance decline across all classes. MiB and PLOP leverage multiaspect knowledge distillation to alleviate forgetting on old semantics, but they remain vulnerable to semantic chaos in fine-grained tasks, especially in long-step 15-2 and 10-5 settings, where subtle interclass differences amplify drift and misclassification. Recent continual semantic segmentation models, including IDEC and LAG, emphasize the anti-forgetting via contrastive learning, which alleviates the semantic confusion among fine-grained classes. The proposed model achieves the highest mean PQ of all classes over multiple tasks. Particularly, FineCL outperforms CPP by 0.90%, 0.53%, and 0.98% in PQ on 20-5, 15-5, and 15-2 tasks, respectively, and achieves up to 6.6% relative improvement in the long-step 15-2 setting. While for the 10-5 task, FineCL maintains the highest PQ on the initial classes, demonstrating superior robustness under long-step CL.

2) DroneDeploy: We also validate our model in high-resolution aerial remote sensing images. As reported in Table II, we measure performance on both the initially learned classes and the incremental classes at the final incremental step. Due to the relatively clear semantic boundaries among DroneDeploy categories, recent continual semantic segmentation baselines such as IDEC and LAG achieve competitive accuracy in short-horizon settings like the 20-5 task. However, their performance on newly introduced classes is less satisfactory, indicating limited plasticity when adapting to incremental semantics. In contrast, the proposed FineCL shows advantages in long-step CL tasks like 4-1 and 2-2-2-1, where accumulated

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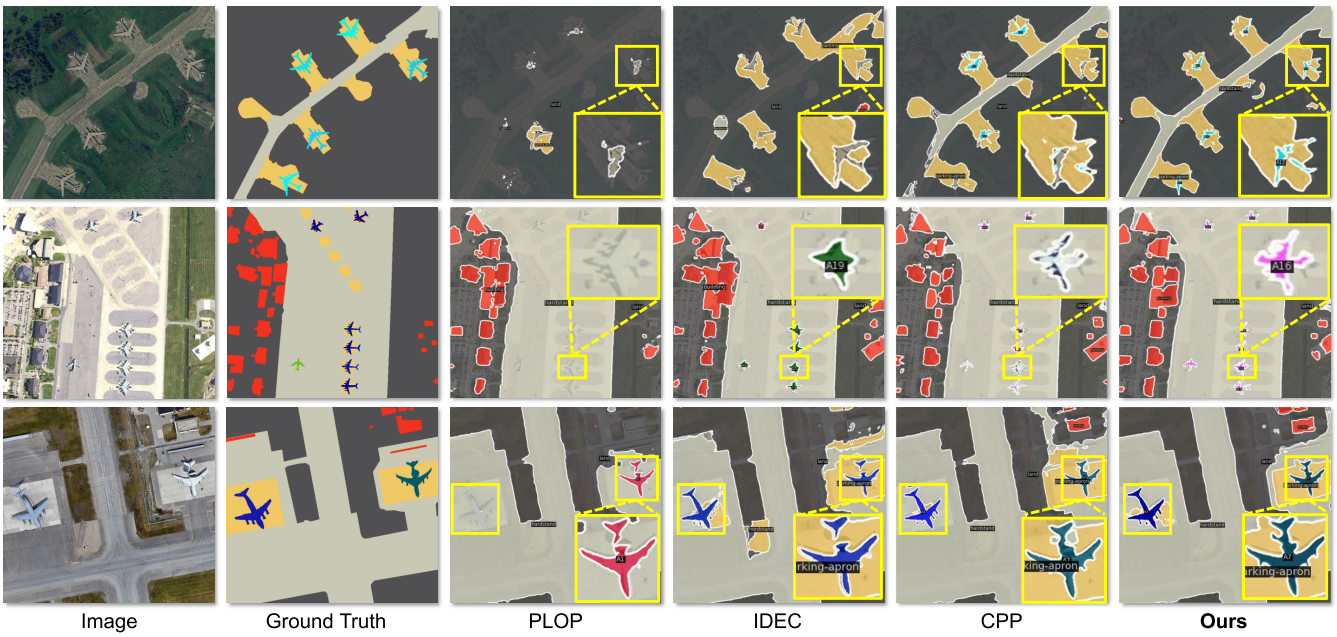


Fig. 4. Qualitative comparison on the FineGrip dataset under the 15-5 setting. Yellow boxes highlight fine-grained instances for detailed inspection.

drift typically leads to larger performance degradation. Specially, FineCL achieves 0.22%, 0.13%, and 0.92% PQ surpassing the current state-of-the-art model CPP on 4-3, 4-1, and 2-2-2-1 tasks, respectively, demonstrating improved stability-plasticity balance under exemplar-free CL.

D. Qualitative Analysis

As depicted in Fig. 4, we qualitatively compare the proposed model with three up-to-date continual segmentation models. All visualizations are generated using models from the final incremental step. The first row depicts a *multiple instance* scenario, where several instances of the same class appear in a single scene. The distillation-based method PLOP suffers from severe catastrophic forgetting and produces incorrect semantic predictions. IDEC preserves stuff semantics but fails to accurately segment foreground thing instances. CPP also struggles with fine-grained thing recognition under IL. FineCL successfully preserves coherent stuff regions while producing accurate and complete thing instances. The second row shows *complex background* scenario. Existing continual segmentation methods exhibit semantic confusion, misclassification, and noticeable forgetting in cluttered scenes. By contrast, FineCL delivers higher segmentation accuracy and semantic recognition performance, particularly in regions where background semantics change substantially across steps. The third row presents *fine-grained* scenario, where the foreground classes have similar semantics. PLOP fails to segment parts of the foreground due to forgetting, while IDEC and CPP maintain better mask integrity but frequently misclassify the fine-grained semantics. The proposed model achieves both segmentation and recognition accuracy, demonstrating stronger robustness in fine-grained CL settings.

Fig. 5 depicts the visualization results on FineGrip across four CL tasks. We visualize representative scenes under four incremental settings, covering three challenging cases, including multiple instances, multiple classes, and fine-grained semantics, to comprehensively assess the model's robustness. Overall, FineCL consistently produces coherent stuff regions and well-separated thing instances. In particular, FineCL remains effective in distinguishing visually similar fine-grained classes, highlighting its strong anti-forgetting capability under long-horizon CL.

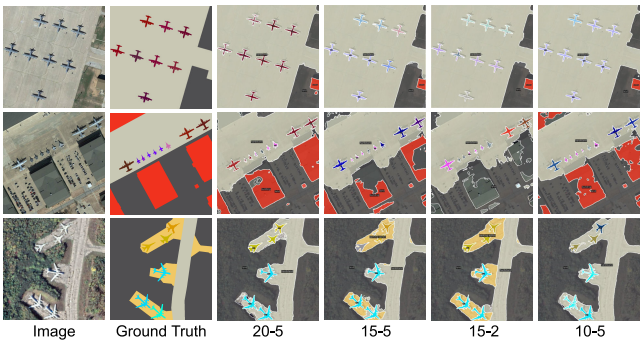


Fig. 5. Visualizations of various CL tasks in FineGrip dataset.

Fig. 6 also analyzes the semantic segmentation results on DroneDeploy. We report qualitative comparisons at the final incremental step to examine the model behavior under challenging aerial scenes with complex layouts and scale variations. Overall, FineCL produces cleaner class boundaries and more coherent region predictions, while effectively reducing misclassification and fragmentation that commonly occur in CL.

E. Ablation Study

1) Module Contribution: Table III reports an ablation study on the contribution of each module. Specifically, when UGPL

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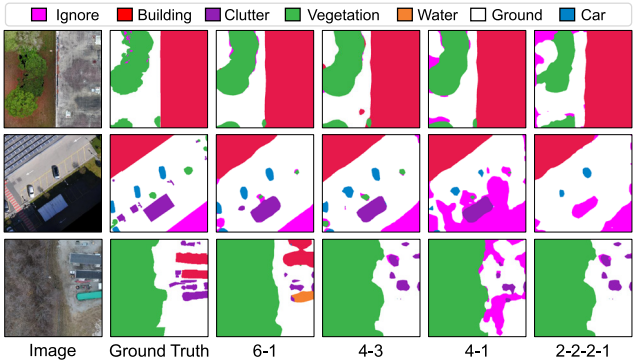


Fig. 6. Visualizations of various CL tasks in DroneDeploy dataset.

TABLE III

INDIVIDUAL CONTRIBUTION OF PROPOSED MODULES. THE RESULTS COME FROM FINEGRIP 15-5 TASK

Module	FineGrip 15-5 $C^{1:15\&st}$	$C^{16:20}$	all
fine tuning	2.85	5.11	3.75
+APE	23.97	19.24	22.08
+TPR	29.54	23.79	27.24
+AKI	31.07	25.49	28.84
+UGPL	31.26	25.53	28.97

is removed, pseudo labels are generated using a fixed confidence threshold ($\tau = 0.5$). APE module utilizes the old model to generate an anchor region embedding to alleviate catastrophic forgetting. It largely improves the PQ than that of fine-tuning. The TPR module introduces triplet prototype ranking to better preserve fine-grained semantic structure, improving both retention on the initial classes and adaptation to newly introduced classes. The AKI module further enhances performance by injecting relevant historical knowledge at each incremental step, improving stability under long-horizon learning. UGPL refines pseudo labels and reduces optimization errors caused by incorrect predictions.

2) Anti-Forgetting Analysis: For the CL task, the catastrophic forgetting typically leads to a monotonic degradation of model performance as new classes are introduced. Thus, analyzing the performance evolution across incremental steps provides a direct indicator of a method's stability and knowledge retention on previously learned categories. As shown in Fig. 7, all methods suffer from conspicuous performance drops from the initial step to subsequent steps, highlighting the difficulty of maintaining fine-grained panoptic semantics. Among the compared methods, PLOP exhibits the most severe degradation, indicating that distillation alone is insufficient to preserve dense fine-grained semantics. IDEC and CPP alleviate forgetting to some extent but still show substantial performance decline. In contrast, FineCL consistently achieves the highest PQ at each incremental step. This trend indicates that FineCL not only preserves old knowledge more effectively but also maintains better overall segmentation quality after multiple incremental updates.

3) Impact of Hyperparameters: The hyperparameter λ defined in 18 controls the strength of the knowledge injection.

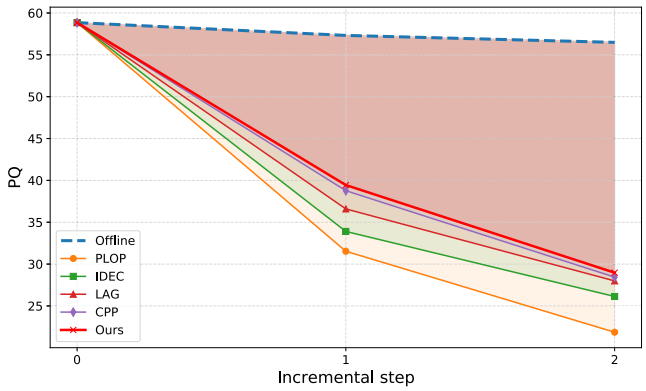


Fig. 7. Comparison of PQ performance on FineGrip 15-5 across incremental steps. The filled area represents the performance gap between the CL model and the offline model.

TABLE IV

IMPACT OF λ ON FINEGRIP 15-5 TASK

λ	0.1	0.5	1.0	1.5	2.0	2.5	3.0
PQ (%)	27.63	27.91	28.42	28.85	**28.97**	28.79	28.72

TABLE V

IMPACT OF PROTOTYPE NUMBER ON FINEGRIP 15-5 TASK

K	1	2	3	5	7	10
PQ (%)	26.22	27.62	28.35	**28.97**	28.92	28.95

Table IV summarizes the results under different λ values, where the best performance is achieved at $\lambda = 2.0$. When λ is too small, the influence of injected historical prototypes becomes marginal, and the model cannot sufficiently compensate for representation drift, resulting in weaker retention of old classes. When λ is overly large, excessive injection may constrain adaptation and introduce interference, leading to a slight performance drop. Overall, the results indicate that a moderate injection weight provides the best stability–plasticity tradeoff, and we therefore set $\lambda = 2.0$ in all experiments.

We also analyze the prototype number K in (7). Table V shows the ablation study on FineGrip 15-5 task. As K increases from 1 to 5, the PQ consistently rises from 26.22% to 28.97%, demonstrating that fine-grained continual segmentation benefits from class-wise multiprototype memory. Nevertheless, more number of prototypes per class yields only marginal changes in PQ, indicating that the benefit gradually saturates once the major semantic modes have been covered. Considering both performance and efficiency, we choose $K = 5$ as the default setting.

4) Impact of Baseline: The pioneers [8], [30] have investigated how different baselines and backbones affect dense prediction performance. Following the protocol, we choose two baselines, including MaskFormer and Mask2Former, with a CNN backbone and a Swin transformer backbone. The experimental results are shown in Table VI.

On the FineGrip 15-5 task, strengthening the baseline architecture yields clear gains. With the same ResNet-101

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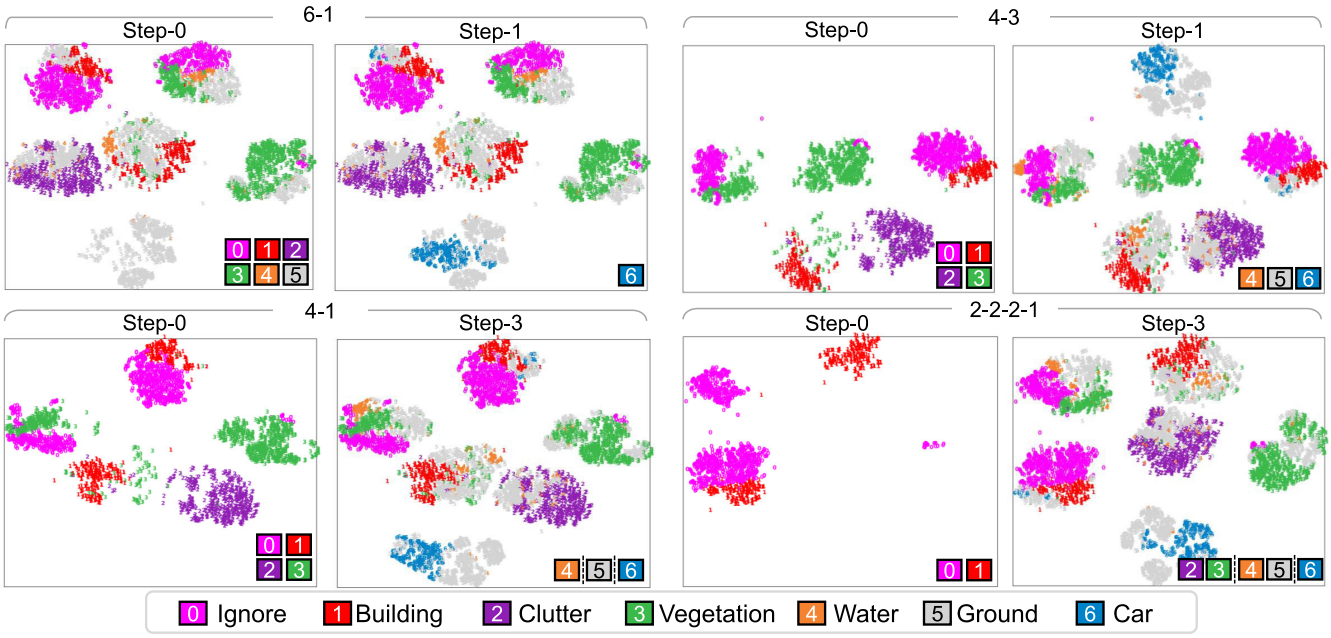


Fig. 8. t-SNE visualizations of four CL tasks on DroneDeploy dataset. It shows the semantic distribution at the initial step and the final incremental step. The dashed lines indicate the set of classes introduced at each incremental step.

TABLE VI
CONTINUAL SEGMENTATION PERFORMANCE USING VARIOUS BASELINES AND BACKBONES

Task	Model	Backbone	$C^{init.}$	$C^{inc.}$	all
FineGrip 15-5	MaskFormer	R-101	25.73	22.19	24.31
	MaskFormer	Swin-T	25.32	23.41	24.56
	Mask2Former	R-101	**31.26**	25.53	28.97
	Mask2Former	Swin-T	30.83	**26.70**	**29.18**
DroneDeploy 4-1	MaskFormer	R-101	54.21	33.90	45.51
	MaskFormer	Swin-T	54.53	31.47	44.65
	Mask2Former	R-101	**56.32**	37.26	48.15
	Mask2Former	Swin-T	55.97	**38.09**	**48.31**

backbone, the Mask2Former-based model outperforms its MaskFormer counterpart by 5.53% PQ on the initial classes and 3.34% PQ on the incremental classes, indicating that a stronger baseline provides a more robust decoding and feature aggregation scheme for continual panoptic learning. Meanwhile, under the same Mask2Former baseline, Swin-T demonstrates a stronger ability to absorb new semantics, leading to more favorable performance on newly introduced categories and reduced degradation under long-horizon updates. For the DroneDeploy 4-1 task, the stronger baseline also shows improved adaptability. While the transformer poses better long-term context representation ability, the anti-forgetting performance on the initial classes is improved by 2.11% mIoU, indicating that enhanced contextual representation can effectively stabilize previously learned semantics in long-horizon continual segmentation.

5) Computational Efficiency: AKI introduces additional computation due to the lightweight projections and Top-K prototype retrieval. As shown in Table VII, we report model parameters, FLOPs, and PQ on FineGrip 15-5 task. Compared with the model without AKI, the model with AKI increases 0.2M parameters and 5.4G FLOPs. While AKI improves 0.51% PQ, which indicates AKI achieves a favorable tradeoff between accuracy and efficiency by enhancing fine-grained knowledge retention.

TABLE VII
COMPUTATIONAL EFFICIENCY OF AKI ON FINEGRIP 15-5 TASK

Model	Param. (M)	FLOPs (G)	PQ (%)
Ours w/o AKI	67.3	315.6	28.46
Ours w/ AKI	67.5	321.0	28.97

6) Interpretability Analysis: To reveal how the semantic distribution evolves across incremental steps, we analyze the feature space learned during continual training. As depicted in Fig. 8, we report t-SNE visualizations on DroneDeploy, where each point corresponds to a sampled pixel embedding and colors denote semantic classes. Overall, FineCL maintains old-class semantic clusters while enabling clearer separation for incremental classes, suggesting the stability under exemplar-free CL. While for challenging long-step CL tasks, including 4-1 and 2-2-2-1, FineCL achieves both the anti-forgetting stability on old classes and the plasticity on the incremental classes. It demonstrates the robustness of FineCL in maintaining a stability-plasticity balance.

7) Robustness to Class Incremental Orders: To evaluate the robustness of the proposed method to different class-incremental orders, we conduct experiments on FineGrip 15-5 under five distinct class sequences, including one ascending order and four randomly permuted orders, as listed below. All the stuff classes are learned at the initial step

$$a: \{[21-25, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10] \\ [11, 12, 13, 14, 15], [16, 17, 18, 19, 20]\}$$

$$b: \{[21-25, 5, 7, 8, 9, 12, 14, 15, 16, 19, 20] \\ [1, 2, 4, 11, 13], [3, 6, 10, 17, 18]\}$$

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TABLE VIII
PERFORMANCE ON VARIOUS CLASS INCREMENTAL ORDERS ON FINEGRIP 15-5 TASK IN TERMS OF PQ (%)

order	$C^{0:15\&st}$	$C^{16:20}$	all
a	31.26	25.53	28.97
b	29.47	27.42	28.53
c	30.43	23.89	27.81
d	32.04	24.65	29.08
e	28.20	26.37	27.47
avg.±std.	30.28±1.35	25.57±1.24	28.37±0.63

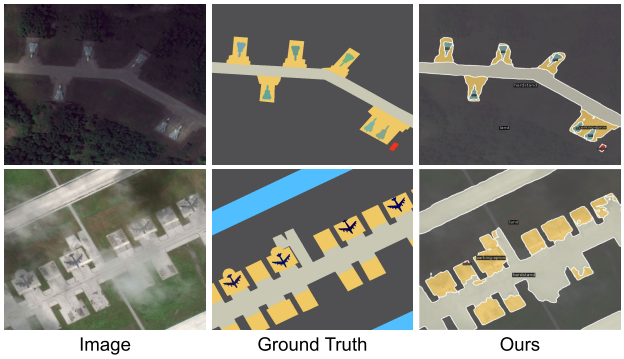


Fig. 9. Robustness analysis of the proposed model on FineGrip 15-2 task.

$$
\begin{aligned}
c : \{&[21-25, 3, 4, 5, 8, 9, 13, 15, 17, 19, 20] \\
&[7, 10, 11, 16, 18], [1, 2, 6, 12, 14]\} \\
d : \{&[21-25, 1, 2, 3, 11, 12, 14, 15, 16, 18, 20] \\
&[7, 8, 10, 17, 19], [4, 5, 6, 9, 13]\} \\
e : \{&[21-25, 2, 3, 5, 7, 9, 10, 12, 13, 14, 19] \\
&[1, 4, 8, 16, 17], [6, 11, 15, 18, 20]\}.
\end{aligned}
$$

For each order, we follow the same training protocol and report the average performance over the final incremental step. Table VIII summarizes the results in terms of the mean performance and standard deviation across the five orders. We evaluate the PQ of $C^{1:10\&21-25}$ and $C^{11:20}$, respectively, to reveal the robustness against different class arrival sequences. We can observe that FineCL exhibits stable performance across orders, achieving an average PQ of 30.28% on $C^{0:15\&st}$, while 25.57% PQ on $C^{16:20}$, and 28.37% on all classes. The relatively small variance, especially for the overall PQ, indicates that our method is insensitive to class incremental orders, consistently preserving old knowledge while maintaining competitive adaptability to new classes.

8) Boundary Robustness Analysis: We further assess FineCL under challenging conditions, as illustrated in Fig. 9. The first row presents a low-light scenario, where the image is overall low contrast and semantic boundaries are visually ambiguous. FineCL preserves smooth and coherent boundaries for the stuff region and maintains accurate recognition and segmentation of small foreground instances, demonstrating strong robustness and generalization of the proposed model. In the second row, under complex weather disturbances such as haze and cloud occlusion, FineCL maintains reliable background semantics with consistent boundaries, but misclassification mainly occurs on foreground instances. We attribute this limitation to degraded recognition of fine-grained foreground semantics under adverse conditions, which may be mitigated by stronger feature backbones or more diverse training data.

VI. CONCLUSION

This article investigates CL for fine-grained panoptic segmentation. We propose FineCL, a unified exemplar-free framework that eliminates the need for storing past training data while effectively mitigating catastrophic forgetting and semantic drift. Multiple remote sensing benchmarks and CL protocols demonstrate that FineCL consistently improves both retention of previously learned classes and adaptation to newly introduced classes, achieving state-of-the-art performance in long-step CL scenarios. However, fine-grained continual panoptic segmentation remains highly challenging under long-horizon class increments, where subtle interclass differences and accumulated distribution shifts can still lead to residual confusion, where subtle interclass differences and accumulated distribution shifts can still lead to residual confusion.

In future work, we plan to extend FineCL to more diverse CL settings, such as domain-incremental and task-agnostic streams, and to investigate more efficient memory mechanisms and an endogenous continuous learning architecture for further robustness in real-world remote sensing applications.

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